

AI & Machine Learning

Visual Study Guide

From Intelligence to Math Foundations

11 Topics | Personal Study Guide | March 2026

1. What is Intelligence?

LEARN

Get better from
past experience

UNDERSTAND

Make sense of
new situations

DECIDE

Choose the best
action

ADAPT

Change behavior
when things change

Dark clouds -> remember rain -> walk faster -> carry umbrella
That process = Intelligence!

2. What is AI?

HUMAN INTELLIGENCE

- Brain learns from experience
- Understands context
- Makes decisions
- Adapts to change

ARTIFICIAL INTELLIGENCE

- Software learns from data
- Finds patterns
- Makes predictions
- Improves over time

AI = Brain (Software) | Robot = Body (Hardware) | Most AI has NO body!

3. Types of AI

NARROW AI

One thing really well | EXISTS

Google Maps, ChatGPT

GENERAL AI

Everything a human can | NOT YET

Science fiction

SUPER AI

Better than ALL humans | SCI-FI

Ultron, Skynet

4. Traditional Programming vs ML

TRADITIONAL

RULES + DATA --> ANSWERS

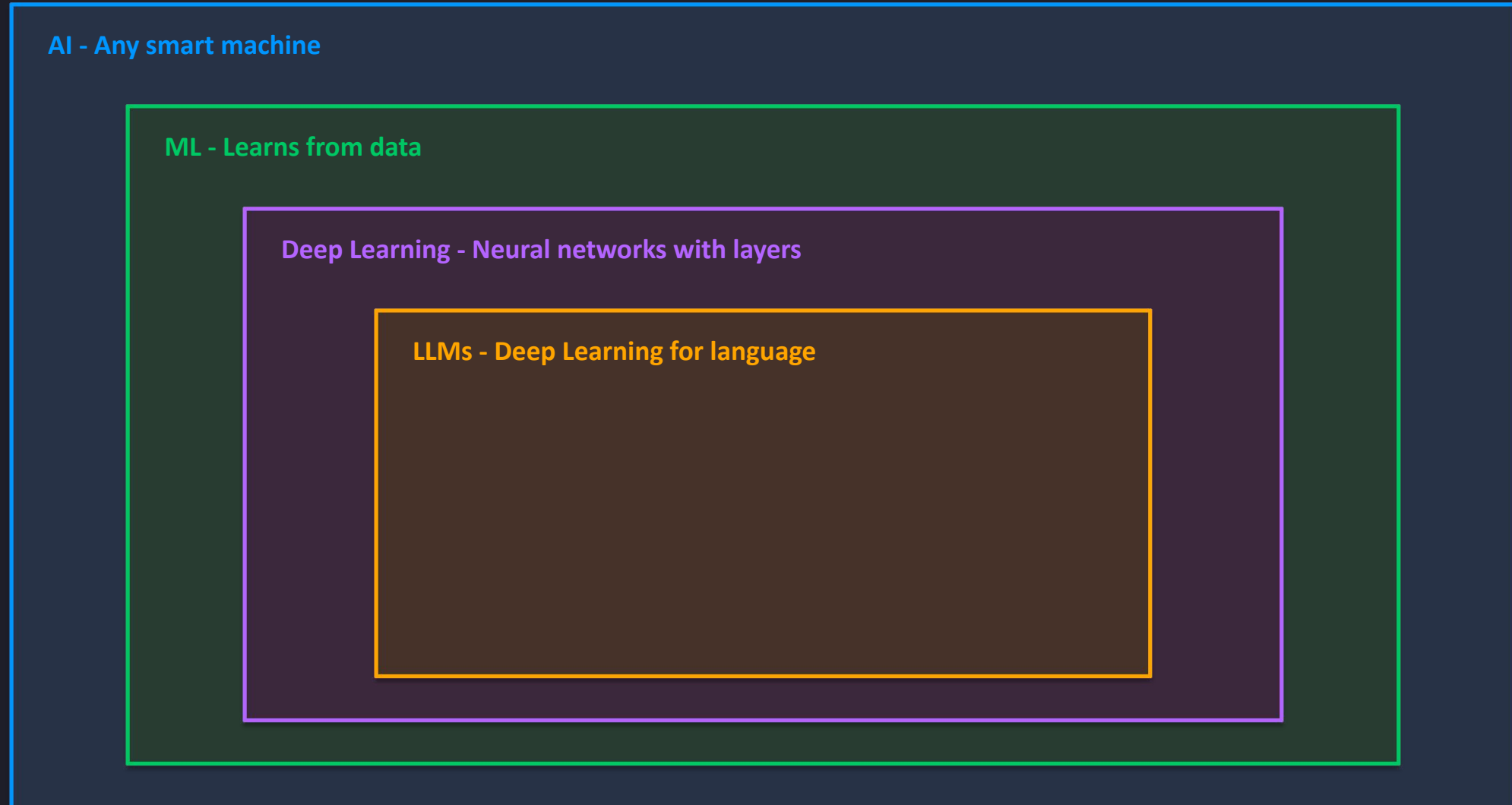
Human writes all rules
Computer follows blindly
Like a recipe book

MACHINE LEARNING

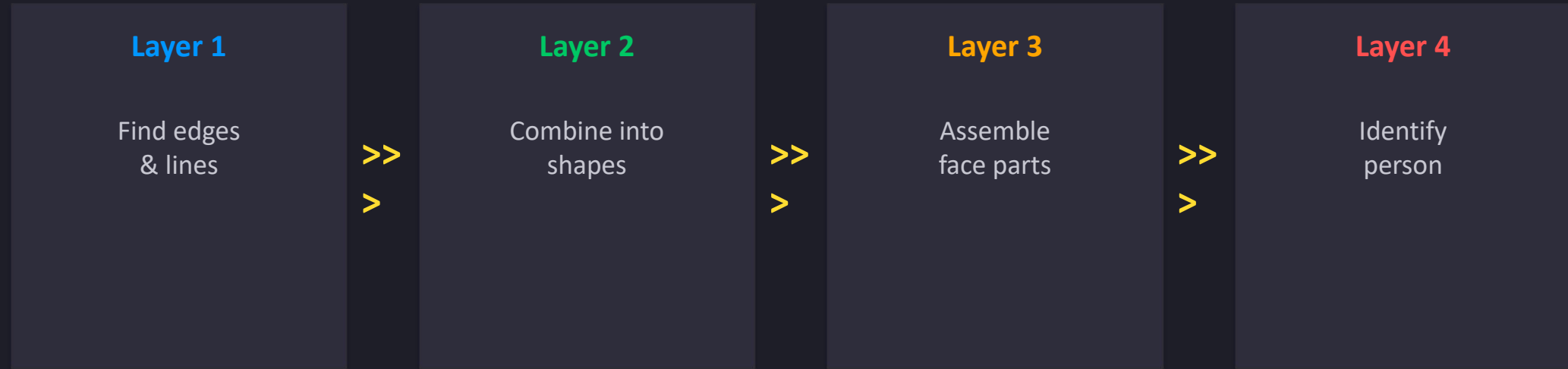
DATA + ANSWERS --> RULES

Computer learns rules itself
Finds patterns from examples
Like tasting 1000 dishes

5. The AI Hierarchy



How Deep Learning Works



Each layer builds on previous: Simple -> Complex -> Answer. "DEEP" = many layers!

6. Supervised Learning

CLASSIFICATION = WORD

Spam? -> Yes/No
Animal? -> Cat/Dog/Bird
Broken? -> Yes/No
Size? -> S/M/L/XL

Like fruit sorting machine

REGRESSION = NUMBER

House price? -> Rs 45 lakhs
Temperature? -> 33 deg C
Delivery? -> 2.5 hours
Marks? -> 87/100

Like mango shopkeeper

"Will student pass?" = Classification | "What marks?" = Regression

Famous Classification Models

Logistic Regression

Line between groups

Fence: apples vs oranges

Decision Tree

Yes/No flowchart

Playing 20 Questions

Random Forest

100+ trees vote

100 doctors majority wins

SVM

Best boundary max gap

Wall with max space

KNN

K nearest neighbors

Rich neighbors = expensive

Famous Regression Models

Linear Regression

Straight line through data

Size vs price line

Polynomial Regression

Curved line

Life isn't always straight

Random Forest Regression

100+ trees average

Classification but numbers

Neural Networks

Many layers complex

Most powerful needs data

7. Unsupervised Learning

CLUSTERING

"Group similar things"

Customers: budget/luxury/deal hunters

Spotify: songs by mood

Google Photos: faces grouped

School Photo: sort 50 photos
with NO names by similarity

ANOMALY DETECTION

"Find the weird one"

Credit card: Rs 500K in Tokyo

Factory: 1000 -> 1500 RPM

Student: 40% -> suddenly 100%

Clustering = who belongs?

Anomaly = who DOESN'T belong?

8. Reinforcement Learning

We define WHAT success looks like. Model discovers HOW.

AGENT

The learner

You / Batsman

ENVIRONMENT

The world

Road / Pitch

REWARD

Good feedback

Balanced! / Four!

PENALTY

Bad feedback

Fell / Got out

Chai: "Good = happy. Bad = face. Figure it out." After 50 attempts = perfect!

Traditional: define WHAT + HOW | RL: define only WHAT, model discovers HOW

How RL is Set Up

1. BUILD

Engineer builds
environment

>

>

2. ACTIONS

Define what agent
can do

>

>

3. REWARDS

Define good
and bad

>

>

4. START!

Model learns
on its own

3 Types of ML Compared

SUPERVISED

"WHAT is this?"

Data WITH labels

Cricket:

Ball, bat, bowler

UNSUPERVISED

"What PATTERNS?"

Data NO labels

Cricket:

Fielding positions

REINFORCEMENT

"HOW to do well?"

Trial + feedback

Cricket:

Hit or defend?

Recognize | Discover | Strategize

Real Project: Exam Monitoring Bot

All 3 types working together!

SUPERVISED

Recognize cheating

Label 10K clips
cheating/normal
Model learns what
cheating looks like

UNSUPERVISED

Spot unusual behavior

Learns normal patterns
Flags anything unusual
even never seen before

REINFORCEMENT

When to alert

Correct alert = +10
False alarm = -10
Learns WHEN to flag

Self-Driving Car: Emergency Handling

Evaluates ALL Options

Brake? Swerve left? Right?
Brake + slight swerve?

Picks LEAST total harm
In 0.001 seconds

Priority System

Human life: -1000 pts
Human injury: -500 pts
Car damage: -30 pts

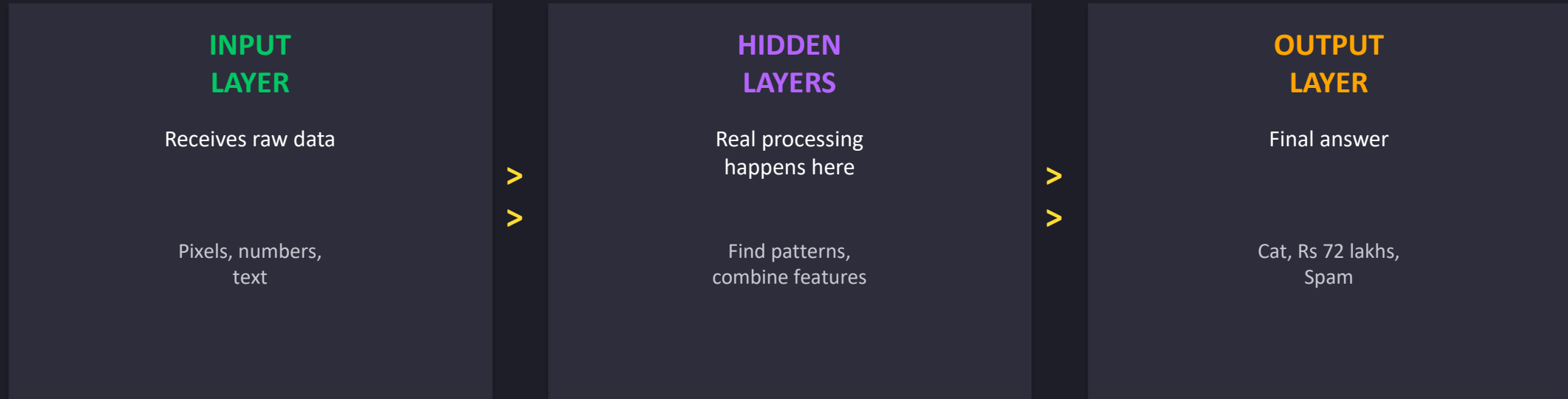
ALWAYS saves lives first!

AI is SAFER: 0.001 sec reaction vs human 1-2 sec panic

PREVENTS emergencies: detects child 5 sec BEFORE, starts slowing

9. Neural Networks

Company Analogy: Reception -> Departments -> CEO



Epoch

One complete pass through
ALL training data

Activation Function = Bouncer

At EVERY neuron: "Important
enough to pass forward?"

10. Math Foundations - Why ML Needs Math

SPEAK -> COMPARE -> FIND -> FIX

VECTORS

SPEAK

Convert data to numbers. ML's language.

DOT PRODUCT

COMPARE

How similar are two things?
High = similar.

DERIVATIVES

FIND

Which weight caused the error?
By how much?

GRADIENT DESCENT

FIX

Walk downhill to minimum error.
Adjust weights.

You = Driver (decide what problem) | Math = Engine (calculations) | Python = Car (press buttons)

The Full Training Process

1. VECTOR

Data -> numbers

[1000, 5]

2. PREDICT

Features x weights

= 20 lakhs

3. ERROR

Actual vs predicted

50-20 = 30 off!

4. DERIVATIVE

Find what's wrong

Size weight too low

5. ADJUST

Fix all weights

0.01 -> 0.018

6. REPEAT

Until accurate

500 rounds!

Train/Test Split & Overfitting

TRAIN/TEST SPLIT

80% data -> TRAINING (model learns)
20% data -> TESTING (never seen!)

Computer checks accuracy automatically
No human reads each output!
Even 100 million entries = 2 seconds

OVERFITTING = Biggest Danger

Model MEMORIZES instead of LEARNING

Like student who memorizes practice questions but fails real exam

99% training + 30% test = OVERFITTING!
Bad data = low on BOTH

Testing by ML Type

Supervised:
Computer compares
predictions vs labels
(automatic)

Unsupervised:
Human checks groups
+ math similarity
score

Reinforcement:
Measure performance
win rate, score,
crashes

11. How Math Connects to ML

Real Estate Agent Analogy: Start dumb -> learn from mistakes -> become expert

The 5 Things to Remember

- | | |
|--|--|
| 1. Model starts DUMB | Random weights = random guesses |
| 2. Each round: guess -> wrong -> adjust | Guided by math, not random |
| 3. After 1000s of rounds = EXPERT | Weights = learned intuition |
| 4. Discovers rules NOBODY taught it | Older house = cheaper (negative weight!) |
| 5. Test on unseen data -> if good -> deploy! | 80/20 split, computer checks |

You don't need to DO the math. Python does it in one line of code.

You need to understand the PROCESS: what's happening and why.

Key Terms - Quick Reference

Intelligence	Learn + Understand + Decide + Adapt	Anomaly Detection	Spot the odd one
AI	Machines that learn & decide	Agent / Environment	RL learner / RL world
Narrow AI	One task (ALL current AI)	Reward / Penalty	Good / Bad feedback
ML	Learns rules from data	Vector	List of numbers = data
Deep Learning	Layers for complex patterns	Dot Product	Similarity score
LLM	DL for language (ChatGPT)	Derivative	What went wrong?
Features / Labels	Input data / Correct answers	Gradient Descent	Fix weights step by step
Classification	Predict a WORD	Learning Rate	Step size (don't overshoot)
Regression	Predict a NUMBER	Epoch	One pass through all data
Clustering	Group similar (no labels)	Overfitting	Memorized, can't generalize

12. Problem Understanding & Model Selection

Ask these 5 questions BEFORE touching any model

1. GOAL?

What am I trying to achieve?

Detect cheating

2. DATA?

What type do I have?

Labeled clips + live feeds

3. ANSWER?

Word or Number or Strategy?

Word = cheating or normal

4. SIZE?

How much data do I have?

10K+ clips = DL territory

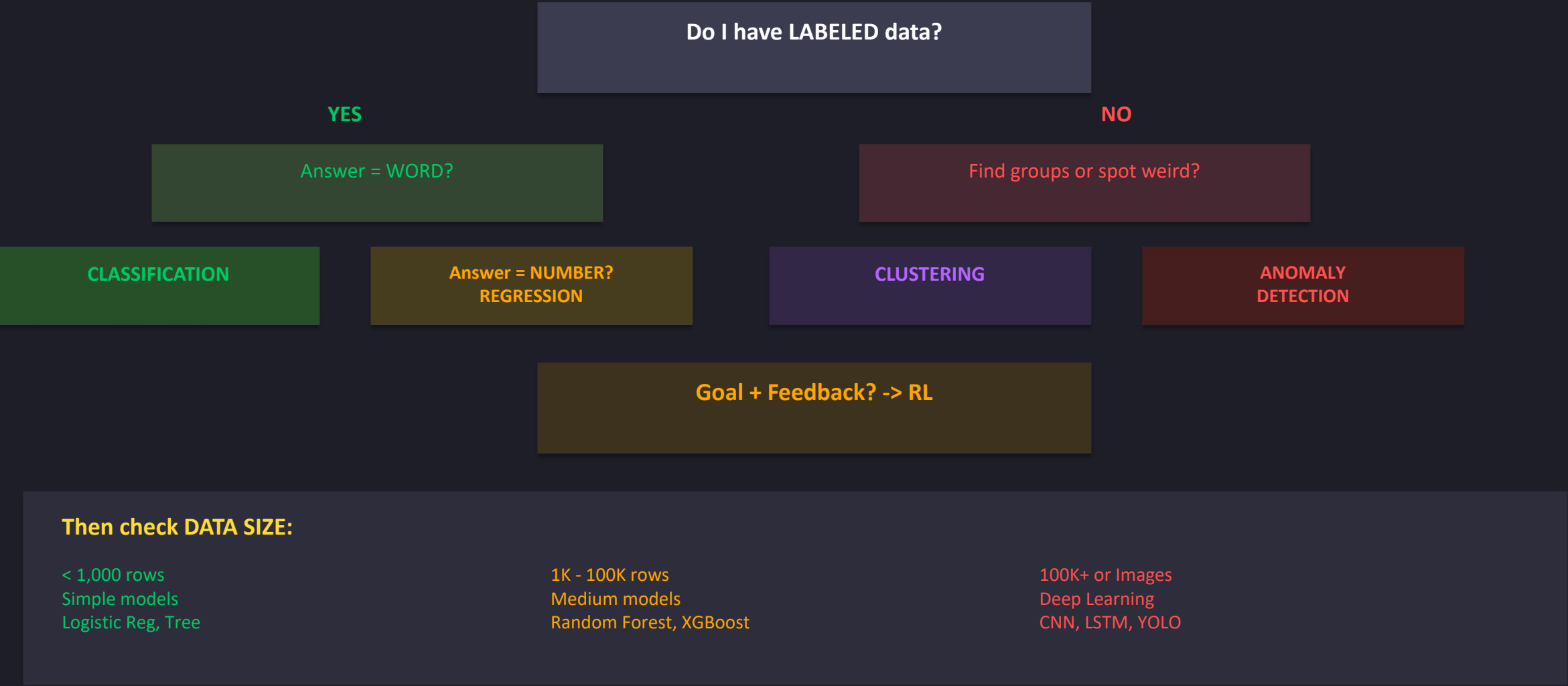
5. SPEED?

Real-time needed?

YES = fast models only

Wrong model = weeks of wasted work. Right questions = right model = faster results!

Model Selection Decision Flowchart



6 Expert Tricks for Model Selection

1. START SIMPLE

Try Logistic Regression first
If low -> Random Forest
Still low -> Neural Network

2. DATA SIZE

< 1K = simple model
1K-100K = Random Forest
100K+ = Deep Learning

3. CHECK EXISTING

YOLO for objects
BERT for text
Most problems = solved!

4. TRANSFER LEARN

Use PRE-TRAINED model
Fine-tune with YOUR data
500 images vs 1 million!

5. MAP SUB-PROBLEMS

Each piece = different model
ExamGuard = 7 models
working together

6. TEST MULTIPLE

Try 3-4 models on same data
Pick best accuracy + speed
Don't guess, MEASURE!

ExamGuard AI - Model Map

Each sub-problem -> right ML type -> best model

Detect Phone

SUPERVISED

YOLO

Pre-trained
real-time

Where Looking?

SUPERVISED

CNN

Best for
images

Cheating?

SUPERVISED

CNN+LSTM

Frames +
time sequence

Unusual?

UNSUPERVISED

Autoencoder

Learns normal
flags rest

Group Behavior

UNSUPERVISED

K-Means

Simple
proven

When Alert?

RL

Custom RL

Balance alert
vs ignore

Track Person

DEEP LEARNING

Re-ID Net

Match across
cameras